

## CPU Sim Facts

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| License type         | Freeware                                                   |
| Author               | Dale Skrien, Department of Computer Science, Colby College |
| Operating system     | Windows, Linux, OSX                                        |
| Latest version       | CPU Sim 3.9.0                                              |
| Disk space           | 7.9 MB                                                     |
| Other requirement(s) | Java runtime environment (JRE) 1.5 or later                |

viewing RAM and registers during execution.

It has several features that make debugging easier. For instance, you can execute the assembled programs either continuously or step-by-step. It allows inspecting or changing any of the values in the register or RAM windows and resuming execution or stepping forward or backward during the execution. If an error is detected during assembling in the program to be run, it is highlighted in the text editor.

## What makes it special

The architecture of CPU Sim is designed at register-transfer level. This means the basic units of the hypothetical CPU are registers, condition bits and memory (RAM). The user does not have to deal with logic gates or transistors on the digital logic level of the machine.

CPU Sim is developed purely using Java Swing package. This makes the program run on any computer with Java runtime environment (JRE) installed in it, irrespective of the platform (Windows, Linux or OSX) it runs on.

When you start CPU Sim, the window comes with a loaded hypothetical CPU known as Wombat1. Here you can assemble and run a program for that architecture. You can also create a new virtual CPU from scratch by going to File > New Machine.

## Why CPU Sim

XSIM Computer Simulator, DLX Simulator, HiRISC Simulator, Simple Computer Emulator and TOM (Thoroughly

## Installation

1. Confirm that Java runtime environment (JRE) 1.5 is installed in the computer. If not, download it (free) from <http://java.sun.com/> (Java comes installed with OSX)
2. If necessary, fix your Path system variable so that it includes the directory installed with Java, in which the Java interpreter application is located. (Not required for OSX)
3. Unzip CPUSim3.9.0.zip file, which produces two folders, CPUSim3.9.0 and MACOSX

### 4. Start CPU Sim

**Windows:** In the folder CPUSim3.9.0, double click on the batch file CPUSim.bat and the program opens automatically. If a window Wombat1 does not open, check again if you have the suitable version of JRE installed in your computer.

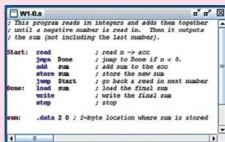
**OSX:** To install the English language version of CPU Sim, double-click on CPUSim3.9.0.app. For Chinese version, double-click on CPUSim3.9.0.Chinese.app file.

### Linux (or for those who prefer using Command Prompt/Terminal Window):

1. Open the command prompt or terminal window and navigate to the folder CPUSim3.9.0.
2. Type the following commands for their respective operating system, in a single line:
  - For Windows:  
java -cp CPUSim3.9.jar;hall.jar;CPUSimHelp3.9.jar cpusim.Main
  - For OSX and Linux:  
java -cp CPUSim3.9.jar;hall.jar;CPUSimHelp3.9.jar cpusim.Main
3. The following four additional optional arguments can be added to the command line, in any order:

- m <machine file name>
- t <text (assembly program) file name>
- l <language>
- c

The -m flag is used to indicate the machine file to be loaded into CPU Sim during start up. If a text file is specified using -t flag, that file is opened during start up. The language for display menus and dialogues can be indicated by the -l file (-l English or -l Chinese). If -l file is not specified, it opens in the default English language. The -c flag (which can be used only together with -t and -m flags) indicates that the Java Swing GUI should not be used, and the corresponding text file and machine file will be loaded and run from the command line. The -p flag is used to specify property files (files that contain information like the preferred fonts, register, RAM windows, and recently opened machine and text files) and is used to load them into CPU Sim during start up.



Sample assembly language

Obedient Moron) Computer Simulator are some of the CPU simulator tools we found online that have features similar to CPU Sim. Though most of these are available for free, they lack many of the features available in CPU Sim. This software is simple enough for self-learning and to work with, yet rich enough to thoroughly understand the concepts and build on them. An

easy user interface and appealing view are additional advantages.

## Support

CPU Sim has a very comprehensive help guide provided in the menu bar (Help > General CPU Sim Help). The installation package also contains a PDF document (CPUSim3.4UserManual.pdf) that explains how various functions are performed using this software. It also includes tutorials with sample assignments that can be run on CPU Sim. Questions and comments can be directly sent to the author Dale Skrien by e-mail. Visit CPU Sim home page [www.cs.colby.edu/djskrien/](http://www.cs.colby.edu/djskrien/) CPUSim for details. ●

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